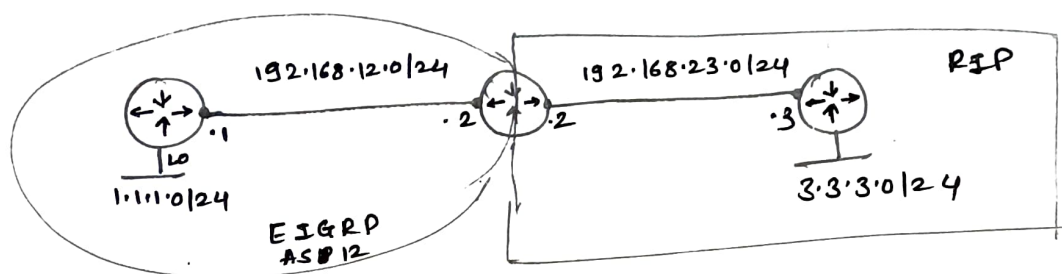


Assignment 4

Q.1



Configure Redistribution?

→ Route Redistribution

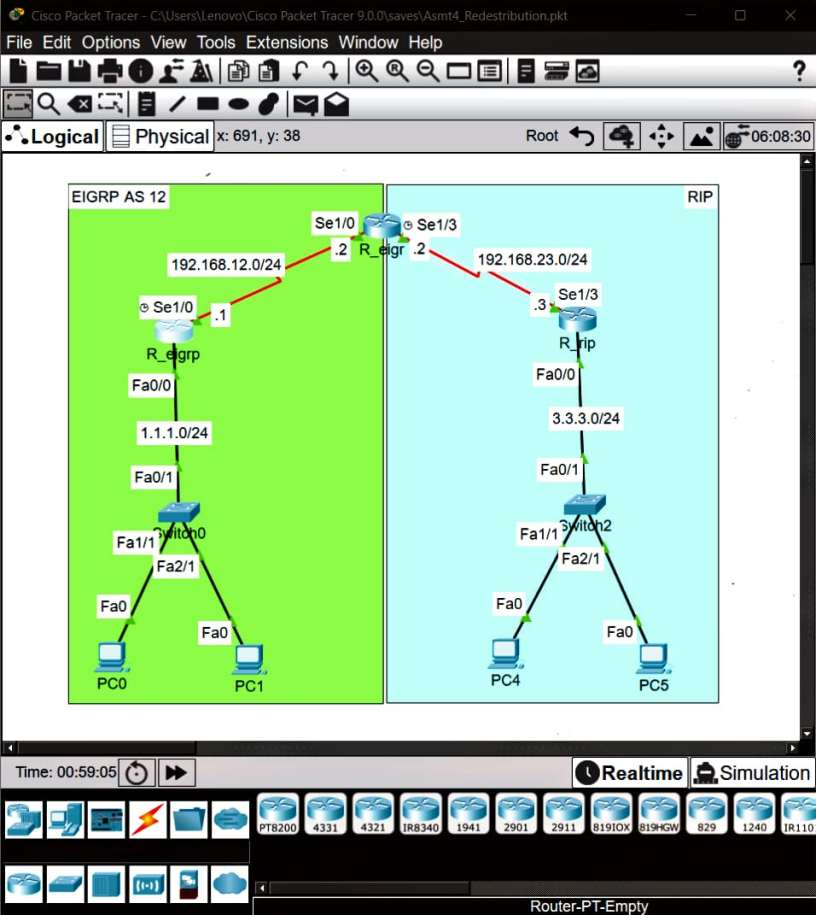
- is a network technique used in routers to exchange routing information between different routing protocols.
- It allows routes learned from one routing protocol to be advertised into another routing protocol, enabling communication between network that use different routing mechanism.

Redistribution Router

↳ Router performs this function mentioned above

↳ It lies at the boundary between two different routing domains.

- Route redistribution widely used in enterprise networks during network migrations, integration of different routing domain, or when multiple routing protocols coexist.



R_eigrp

Physical Config CLI Attributes

IOS Command Line Interface

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

1.0.0.0/24 is subnetted, 1 subnets
C 1.1.1.0 is directly connected, FastEthernet0/0
3.0.0.0/24 is subnetted, 1 subnets
D EX 3.3.3.0 [170/2244096] via 192.168.12.2, 00:02:04, Serial1/0
C 192.168.12.0/24 is directly connected, Serial1/0
D EX 192.168.23.0/24 [170/2244096] via 192.168.12.2, 00:12:15, Serial1/0

Router#
```

R_rip

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

1.0.0.0/24 is subnetted, 1 subnets
R 1.1.1.0 [120/1] via 192.168.23.2, 00:00:06, Serial1/3
3.0.0.0/24 is subnetted, 1 subnets
C 3.3.3.0 is directly connected, FastEthernet0/0
R 192.168.12.0/24 [120/1] via 192.168.23.2, 00:00:06, Serial1/3
C 192.168.23.0/24 is directly connected, Serial1/3

Router(config)#
```

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Top

Q. 2 AD value?

→ ADMINISTRATIVE DISTANCE (AD)

- AD is a numerical value used by routers to measure the trustworthiness of a routing information source.
- When a router learns routes to the same destination network from multiple routing protocols or routing source, AD is used to determine which route should be installed in the routing table.
- * The route with the lowest administrative distance is considered the most reliable and is selected as the best route.
- AD is a local parameter to the router, meaning it only influences route selection on that particular router and not advertised to another router. It helps resolve conflicts when different routing protocols (RIP, OSPF, EIGRP etc.) advertise paths to the same destination. By assigning different AD value to different routing source, router can prioritize routes learned from more reliable protocols.

Route Source	AD
Connected Route	0
Static Route	1
EIGRP (internal)	90
OSPF	110
RIP	120
Unknown or Untrusted Route	255

- Each routing source has default AD value assigned by router OS. For example: direct ~~truste~~ connected is most trusted.

- AD is therefore essential for maintaining stable and efficient routing decision, especially in networks where multiple routing protocols operate simultaneously. It ensures that the router selects the most reliable path and avoid routing conflicts between